DNA of a Hit Song:

A Data-Driven Investigation

DS612 — Interactive Data Visualization

Group — 8

Katrodiya Het (202518005)

Divya Mashruwala (202518042)

Aksh kumar Patel (202518046)

Sanil Shah (202518060)

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Introduction

What makes a song unforgeable—and more importantly, what makes it popular? Every day, thousands of songs are released into the world. Some disappear without a trace, while a select few rise to dominate charts, playlists, and public memory. Is this success driven by measurable musical patterns, or is it simply unpredictable?

This project begins with a simple assumption: that features like danceability, energy, and tempo shape a song's popularity. But instead of accepting this at face value, we put it to the test through a structured, data-driven investigation.

The journey unfolds across four analytical phases, progressing from individual feature analysis to multiple-factor interactions and finally to external commercial influences shaping modern music success.

What emerges is not a formula, but a pattern—a hidden structure we refer to as ***"The DNA of a Hit Song"***. This report reveals that popularity is not the result of a single defining trait, but a complex interplay of balance, interaction, and context.

This is not just an analysis of songs. It is an explora on into why some music resonates—and why most does not.

Narrative Flow: This report follows a structured analytical progression:

- (1) analysing individual features,
- (2) exploring balance in musical attributes,
- (3) studying feature interactions, and
- (4) incorporating external influences to explain real-world popularity.

Context & Background

With the rapid growth of music streaming platforms, millions of songs are available to listeners, yet only a small fraction achieve high popularity. Understanding what makes a song successful has become an important question for artists, producers, and platforms alike.

Musical attributes such as danceability, energy, tempo, and emotional tone are often assumed to influence a song's popularity. However, it remains unclear whether these features alone can explain why certain songs become hits.

Problem Statement

While several musical features are believed to drive popularity, there is no clear evidence that any single attribute can consistently predict a song's success.

This project aims to investigate whether commonly assumed factors such as danceability, duration, and tempo truly influence popularity, and to explore deeper patterns that define a **"hit song"**.

Objectives

- Analyse the relationship between audio features and song popularity
- Test common assumptions about hit songs
- Identify deeper patterns beyond individual features
- Derive the "DNA of a Hit Song" using data-driven insights

Key Questions:

- Can individual features predict song popularity?
- Does balance matter more than extremes?
- Do feature interactions better explain success?
- How significant are external influences?

Visualization Goals

This study leverages both single-frame and multi-frame visualisations to uncover patterns in song popularity. Single-variable plots help identify isolated trends, while multi-dimensional visualisations reveal interactions between features.

Charts Used

- **This study uses:**
  - **Single-frame visualizations:** such as histograms and bar charts to analyse individual feature distributions.
  - **Multi-frame visualizations:** such as heatmaps and interaction plots to capture relationships between multiple variables.

1. **Binned Bar Chart**
   - It groups continuous numerical data into interval (bins) and shows the frequency count of each bin

- **Task:** Understanding distribution patterns and to identify concentration zones
- Marks: 2D
- Channels:
  - Vertical Position
  - Horizontal Position

2. **Histogram**
   - A type of bar chart used to visualize the distribution of a single continuous variable
   - **Task:** to analyze the distribution shape of variables, identify skewness.
   - Marks: 2D
   - Channels:
     - Vertical Position
     - Horizontal Position
3. **Stacked Bar Chart**
   - Shows how a total value is divided across categories
   - **Task:** To analyse genre-wiz contribution to total popularity
   - Marks: 1D
   - Channels:
     - Length
     - Hue
4. **2D Heat Map**
   - A heatmap represents values using colour intensity across two dimensions
   - **Task:** To analyze relationship between two continuous variable
   - Marks: 2D
   - Channels:
     - Saturation, Luminance (Magnitude)
     - Vertical Position
     - Horizontal Position
5. **Line Chart**
   - Connected data point to show trends over a continuous axis.
   - **Task:** To visualize trends and relationships over ordered data
   - Marks: 0D
   - Channels:
     - Hue (Categorical)

  - Vertical Position
  - Horizontal Position

Dataset Description

The dataset used in this study consists of approximately 191,049 songs obtained from Spotify, where each song is represented through a set of quantitative audio features.

Data Overview

- Number of observations: ∼191,049
- Number of features: 10+
- Data type: Structured tabular data

Key Variables

- Danceability: Measure of how suitable a track is for dancing (0–1)
- Energy: Intensity and activity level of a song (0–1)
- Valence: Musical positivity or emotional tone (0–1)
- Tempo: Speed of the song in beats per minute (BPM)
- Loudness: Overall loudness in decibels (dB)
- Acousticness: Likelihood of the track being acoustic
- Duration: Length of the track (in milliseconds)

Target Variable

- Popularity: A score between 0 and 100 representing the song's popularity

Derived Features

To enhance analytical depth, additional transformations and derived features were created

- Duration Category: Songs categorised into short, medium, and long durations
- Feature Binning: Continuous variables grouped into ranges to observe patterns
- Interaction Features: Pairwise combinations such as Energy × Valence and Danceability × Energy

These derived features enabled multi-dimensional analysis and helped uncover patterns not visible through raw features alone. Also, the dataset is high-dimensional and continuous, making it suitable for both statistical analysis and visualisation-driven exploration.

Data Structure and Variable Types

- **Continuous Numerical Variables:** Danceability, energy, tempo and loudness
- **Categorical Variables (derived):** Duration Category, etc. . .

Data Preprocessing

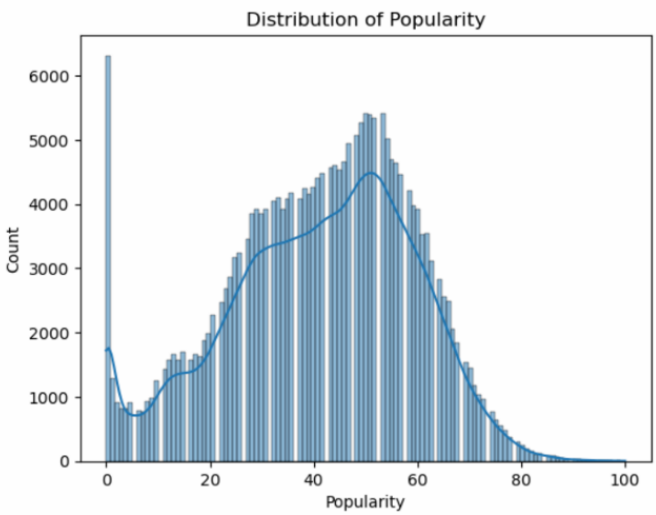
Before analysis, the dataset was cleaned to ensure consistency and accuracy.

Steps

- Removed missing values
- Eliminated duplicate records
- Verified data types and feature ranges

Exploratory Data Analysis (EDA)

- **Popularity Distribution:**



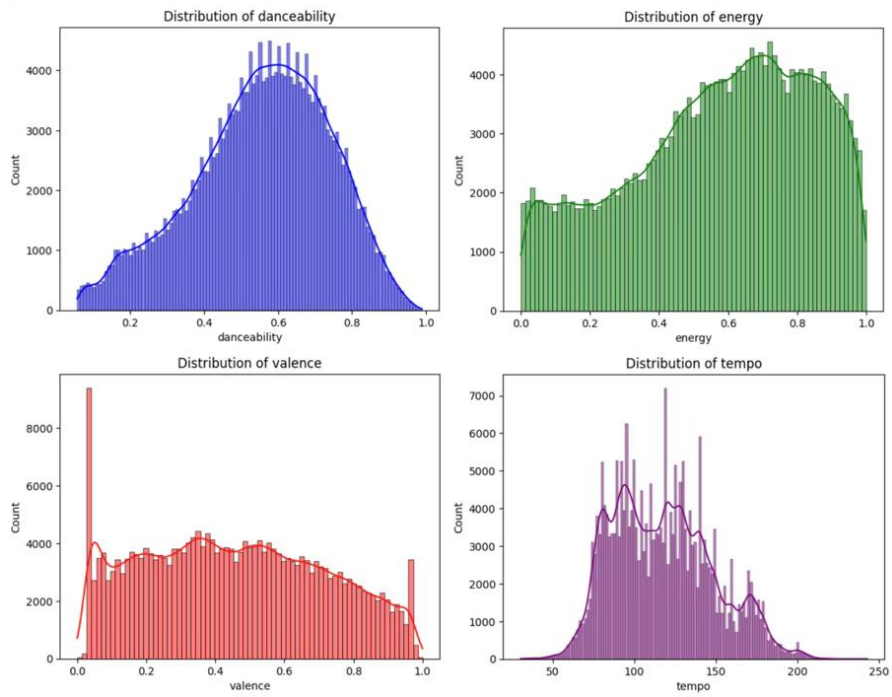
The distribution of popularity shows that most songs fall within a moderate range (40–60), while only a small number achieve very high popularity.

A noticeable spike at zero indicates the presence of songs with no measurable popularity.

Insight: Highly popular songs are rare, suggesting that success is not easily achieved.

- **Feature Distributions:**

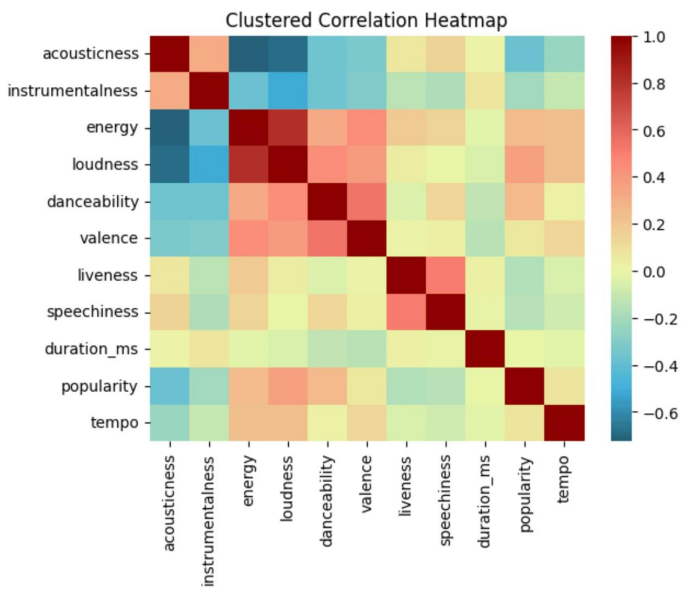
The distributions of key features reveal that:



- Danceability is mostly moderate
- Energy levels are generally high
- Valence is centred around moderate values
- Tempo is concentrated within 80–130 BPM

Insight: Songs follow relatively consistent structural patterns.

- **Correlation Analysis:**



- Danceability, energy, and loudness have weak positive correlations with popularity
- Acousticness has a negative correlation.
- Most features have very weak relationships with popularity.
- A noticeable spike at zero indicates the presence of songs with no measurable popularity.

- Overall, the analysis indicates that no single feature strongly explains popularity. While songs share common structural characteristics, these alone are insufficient to determine success.

  - This observation marks a critical turning point in the analysis. It shifts the investigation from isolated feature evaluation to a structured, hypothesis-driven exploration of deeper patterns and interactions.

Transition

- Motivated by these findings, we adopt a hypothesis-driven approach to test commonly held assumptions about hit songs

Hypothesis 1: "The Search for a Single Formula"

Objective: Test if single features independently drive popularity

- **Q1: Danceability independently influences popularity**

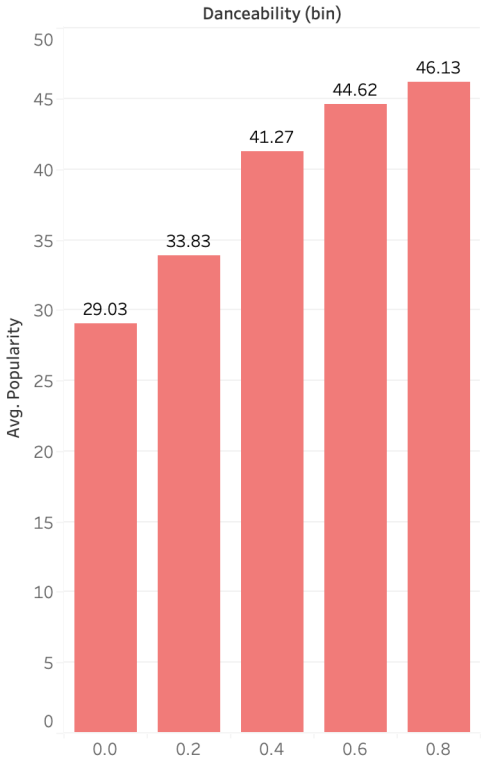


Figure 1. Influence of danceability on popularity

We observe a positive relationship between danceability and popularity. However, the increase is gradual and remains within a moderate range. Even highly danceable songs do not achieve high popularity, indicating that danceability alone is not sufficient to create a hit song.

- So, our hypothesis is supported, but not useful to explain the popularity.

- **Q2: Song duration independently influences popularity**

Songs were categorised into short, medium, and long durations to simplify analysis. The results show that medium-duration songs achieve the highest average popularity, while shorter and longer songs perform relatively worse. This suggests the presence of an optimal duration range.

However, even this peak value is relatively moderate and does not reach high levels of popularity. This suggests that while duration influences popularity, it is not sufficient to create a hit song.

- Thus, our hypothesis is supported, but not useful to explain the popularity.

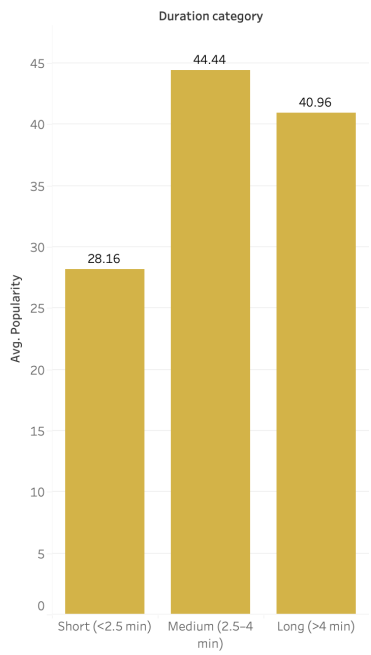


Figure 2. Influence of duration on popularity

- **Q3: Liveness independently influences popularity**

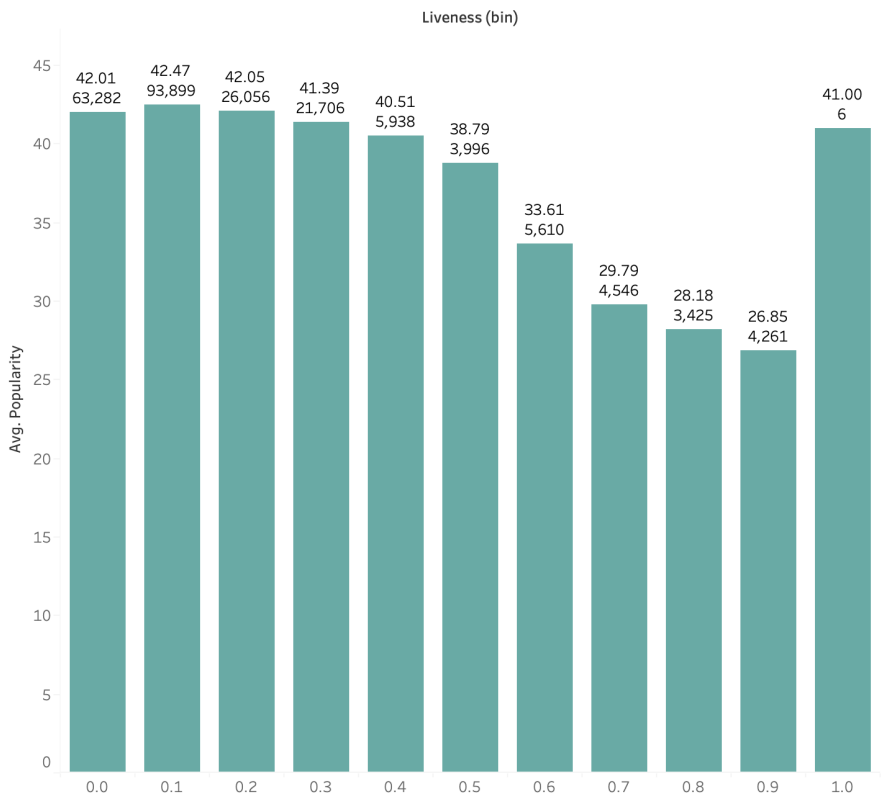


Figure 3. Influence of Liveness on popularity

The analysis of liveness reveals a clear negative relationship with popularity, where songs

with lower liveness values tend to achieve higher average popularity, indicating a general preference for studio-recorded tracks over live recordings. However, this relationship represents a trend rather than a definitive rule. Not all songs with low liveness are highly popular, and some songs with higher liveness still achieve moderate success.

This variation suggests that while liveness influences popularity, it does not fully determine it. Therefore, the hypothesis that liveness impacts popularity is supported, but it cannot be considered the sole driver of success. Popularity is shaped by multiple musical attributes as well as external factors, and liveness should be interpreted as one contributing factor among many.

Narrative: The analysis of individual features reveals that while certain attributes show noticeable trends, none of them independently produce high popularity. Each feature contributes partially, but fails to provide a complete explanation. This indicates that song success cannot be reduced to a single defining characteristic, motivating the need to explore more structured patterns beyond isolated features.

Hypothesis 2: "The Power of Balance"

Objective: Check if moderation (not extremes) leads to success

Q4: Songs follow a common tempo range

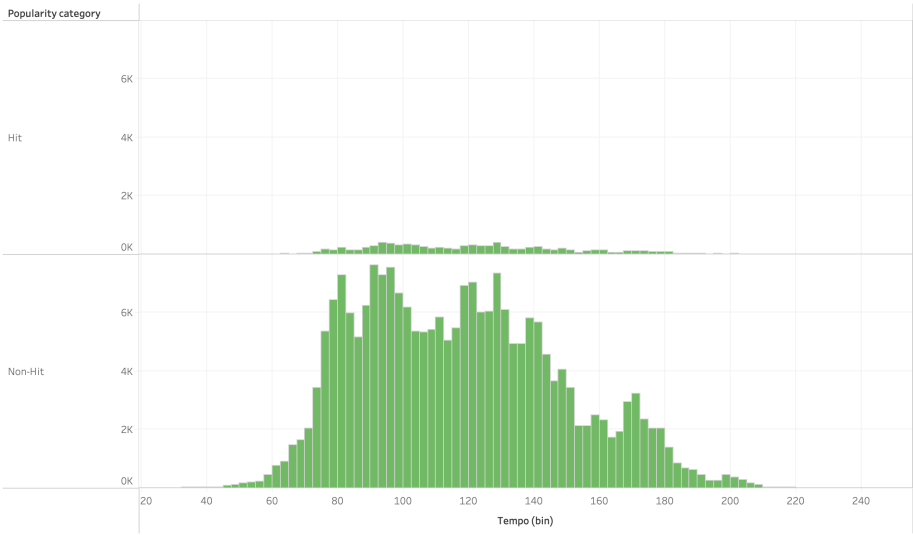


Figure 4. Effect of song tempo range on popularity

The comparison of tempo distributions for hit and non-hit songs shows a similar concentration within the 80–130 BPM range. This suggests that tempo is a standardised feature across songs and does not uniquely differentiate popular songs from others.

So, our hypothesis is supported, but not useful to explain the popularity.

Q5: Balanced energy leads to higher popularity

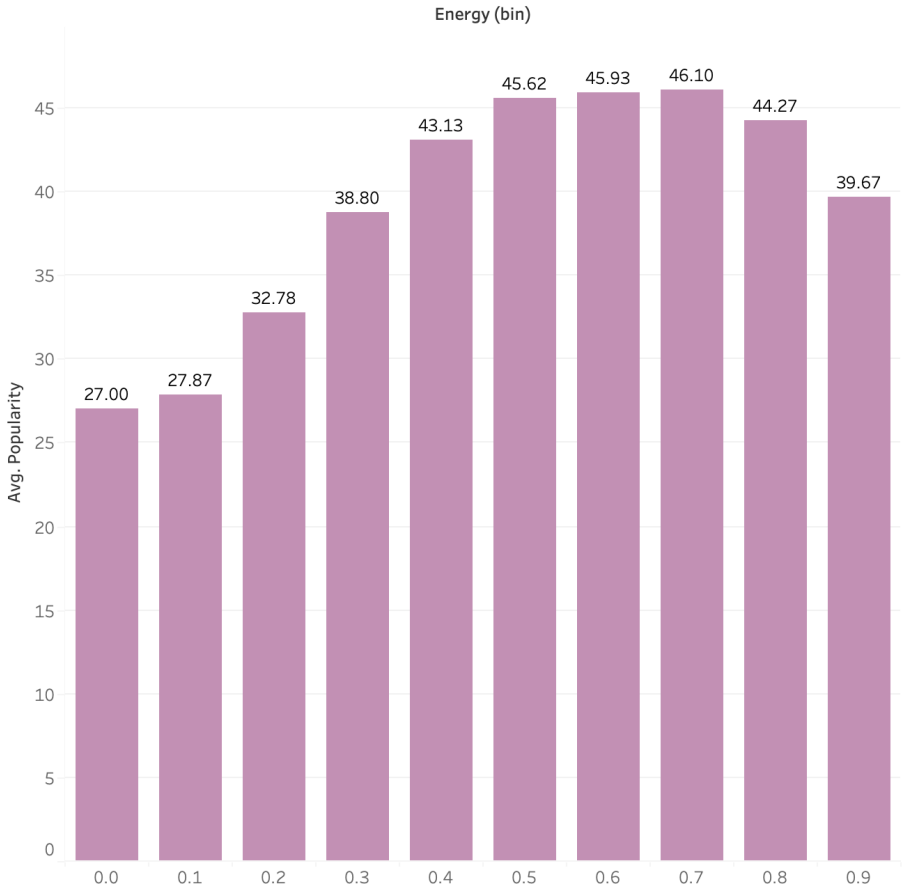


Figure 5. Influence of energy on popularity

The analysis shows a non-linear relationship between energy and popularity. Popularity increases with energy up to a moderate level (around 0.6–0.7) and then declines at higher levels. This suggests that songs with balanced energy are more likely to achieve higher popularity, while extreme energy levels are less effective. Thus our hypothesis is supported. Popularity is maximized at balance, not extremes.

Q6: Moderate valence Performs better

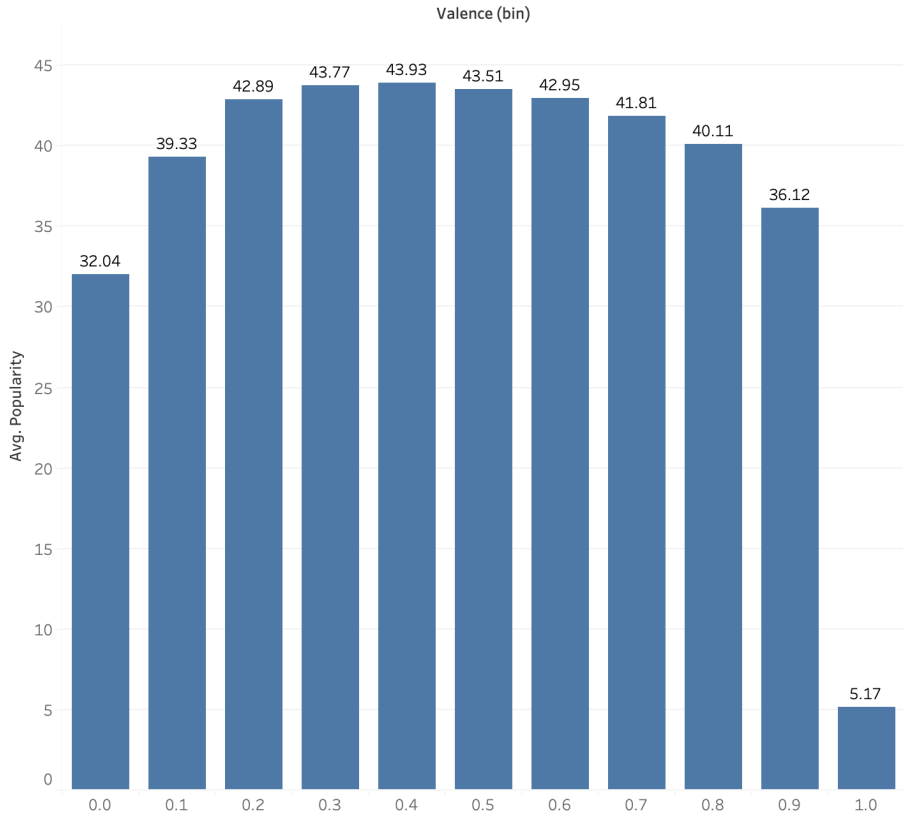


Figure 6. Influence of valence on popularity

The analysis shows that popularity is highest at moderate valence levels, while songs with very low or very high valence tend to have lower popularity. This indicates that balanced emotional tone is more effective than extreme emotional expression. The analysis shows that popularity is highest at moderate valence & energy levels, while songs with very low or very high valence or energy tend to have lower popularity. This indicates that balanced emotional tone is more effective than extreme emotional expression.

Narrative: The findings from this phase suggest that popularity is not driven by extreme values but by balance across musical attributes. Songs with moderate and well-aligned features consistently perform better than those with highly skewed characteristics. However, while balance provides a stronger explanation than individual features, it still does not fully account for variations in popularity, leading to the exploration of feature interactions.

Hypothesis 3: "Synergy: When Features Work Together"

Objective: Understand multi-feature interactions

Q7: Balanced danceability + energy levels drives the popularity of song

Danceability_bins	Energy (bin) 0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.0	29.16	31.12	31.77	30.91	31.32	27.14	34.00	51.33	45.18	30.33
0.1	27.80	27.12	27.57	29.97	31.89	32.94	39.13	40.19	40.55	38.63
0.2	24.84	24.00	26.96	30.72	35.98	38.07	41.44	43.75	42.85	38.52
0.3	26.71	26.07	31.14	36.18	40.23	41.72	44.10	43.68	41.71	37.00
0.4	28.70	29.91	35.05	39.77	42.51	43.47	43.31	44.18	42.43	38.27
0.5	29.00	34.34	37.85	40.33	43.02	44.39	44.27	44.86	43.64	40.34
0.6	24.79	31.50	36.25	40.71	43.36	45.79	45.94	46.52	45.38	41.70
0.7	22.35	25.04	32.95	41.28	45.23	47.74	47.64	47.54	46.46	42.55
0.8	15.60	18.73	26.58	39.78	46.12	48.82	49.17	49.15	45.20	42.84
0.9	0.50	14.81	18.83	34.77	44.42	51.53	53.54	50.43	49.29	46.30

Figure 7. Danceability–Energy Interaction

The interaction between danceability and energy reveals a distinct high-popularity region where both features are moderately high. Songs with low danceability perform poorly regardless of energy levels, while energy alone does not significantly improve popularity. This indicates that popularity is driven by the combined effect of multiple features rather than any individual attribute.

Q8: Vocal-Centric Songs Achieve Higher Popularity

Instrumentalness (bin)	Speechiness (bin) 0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.0	43.33	45.01	48.84	49.31	46.01	38.13	26.38	24.04	19.84	20.95
0.1	40.01	39.75	43.01	38.49	43.29	31.43	19.50	12.50		14.00
0.2	38.43	39.72	45.08	39.95	43.72	35.00	36.00		33.00	
0.3	37.71	40.26	41.24	37.30	34.30	21.50		28.00		10.00
0.4	36.71	33.19	37.35	33.63	32.00	28.00				
0.5	35.98	35.88	36.31	37.74	37.40	29.00	24.00			
0.6	34.65	37.19	36.14	34.04	24.00	39.25	61.00			
0.7	33.64	34.06	39.39	31.32	31.50	18.00	35.00	44.00	0.00	
0.8	32.31	32.92	35.31	31.88	35.87	35.00		35.00		
0.9	31.31	30.05	23.77	33.63	30.09	38.67	31.00	23.00		

Figure 8. Speechiness–Instrumentalness Interaction

Songs with **low speechiness and low instrumentalness** achieve the highest average popularity, indicating that vocal-driven and structured music performs best. As speechiness increases, popularity declines significantly, suggesting that excessive spoken content reduces listener engagement. Similarly, higher instrumentalness leads to lower popularity, highlighting the importance of vocals in mainstream music. Regions with both high speechiness and instrumentalness are sparse, as these features rarely coexist. Some isolated high values are due to low sample sizes and are not reliable. Overall, popularity is highest when both features are low, emphasising the role of balanced, vocal-centric compositions.

Our hypothesis is supported. This indicates that vocal-centric and structured Compositions are more favourable for mainstream success, while excessive spoken Content or purely instrumental elements reduce audience appeal.

Q9: Songs with moderate energy and moderate valence achieve higher popularity

Energy (bin) \ Valence (bin)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.0	27.57	27.37	26.95	26.06	24.35	23.66	21.41	20.53	15.55	15.52	
0.1	27.86	28.46	29.86	30.89	26.93	25.26	22.13	19.59	17.83	15.41	4.00
0.2	31.46	35.17	36.33	36.85	33.09	30.04	28.74	24.85	21.90	14.42	
0.3	37.42	41.83	43.49	42.27	40.48	38.67	34.99	31.70	29.27	19.59	1.00
0.4	42.37	45.87	46.94	46.56	45.31	43.02	41.22	37.91	35.18	27.46	
0.5	43.12	47.50	48.67	48.27	47.75	46.33	45.17	41.54	40.97	35.08	
0.6	43.27	47.02	48.41	47.56	48.00	47.13	46.20	44.65	41.70	38.05	
0.7	41.06	43.91	47.84	47.58	48.03	47.73	46.73	45.28	44.17	40.23	0.00
0.8	35.73	40.58	43.67	44.89	45.21	45.48	45.48	45.50	43.23	40.78	12.50
0.9	32.92	37.19	39.40	40.05	40.53	40.87	40.27	40.81	39.00	38.18	

Figure 9. Energy–Valence Interaction

The interaction between energy and valence reveals a clear pattern where songs achieve higher popularity at moderate levels of both features. Low-energy and low-valence songs tend to perform poorly, while extreme combinations also show reduced popularity. This suggests that a balanced combination of emotional tone and intensity is more effective in achieving higher popularity.

Thus, our hypothesis is supported.

Q10: Song with moderate energy and loudness achieves high popularity

Energy (bin)	Loudness (bin) -55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
0.0	4.00	23.14	30.31	30.88	28.06	26.93	25.59	27.46	31.43	23.97		
0.1				39.00	24.71	24.74	24.72	26.65	33.22	38.65	0.00	
0.2					28.00	25.71	24.74	28.10	34.84	41.07	14.10	
0.3					31.00	23.05	22.39	29.93	38.10	44.59	32.48	
0.4						20.80	21.83	27.99	39.18	47.06	50.45	
0.5						23.67	20.06	26.59	38.24	47.88	52.26	
0.6						38.00	18.60	23.48	35.77	46.84	51.48	
0.7							18.27	24.76	31.73	45.83	50.56	
0.8							17.25	22.54	29.58	42.52	47.72	53.50
0.9						33.00	25.00	22.26	26.61	37.47	41.00	32.95

Figure 10. Energy–Loudness Interaction

The interaction between energy and loudness reveals that songs achieve higher popularity when both features are aligned. Specifically, moderate to high energy combined with higher loudness levels results in the highest popularity. In contrast, mismatched combinations show reduced performance, indicating that alignment between musical intensity and production characteristics plays a key role in song success.

Thus, the hypothesis is supported. Product must match performance energy. Songs need to sound loud to succeed.

Narrative: Moving beyond individual and balanced features, the analysis explores how combinations of attributes influence popularity. Results indicate that songs with well-aligned features outperform those optimised on a single dimension.

Hypothesis 4: "Beyond the Song: Industry Dynamics"

Objective: Understand external influences

Feature Engineering

The previous analyses demonstrated that neither individual features nor simple interactions fully explain song popularity. While patterns of balance and synergy were observed, they remained descriptive rather than measurable. To address this, we construct a quantitative framework that transforms raw audio features into standardised "scores", representing how closely a song aligns with the characteristics of a typical hit.

Each musical attribute is transformed into a normalised score between 0 and 1, where higher values indicate stronger alignment with the hypothesised "hit song characteristics".

1. **Danceability Score:** Measures how suitable a track is for dancing, with higher values preferred.

$$\textbf{danceability_score} = \begin{cases} 1, & \text{if } d \geq 0.65 \\ \dfrac{d}{0.65}, & \text{otherwise} \end{cases}$$

- Highly danceable songs tend to perform better, but beyond a threshold, the benefit saturates.

2. **Energy Score:** Captures how close the energy level is to an optimal value (~0.65).

$$\textbf{energy_score} = \max\left(0,\ 1 - \frac{|e - 0.65|}{0.65}\right)$$

- Extremely low or high energy reduces popularity, while moderate energy maximizes engagement.

3. **Valence Score:** Represents emotional positivity, optimized around ~0.45.

$$\textbf{valence_score} = \max\left(0,\ 1 - \frac{|v - 0.45|}{0.45}\right)$$

- Songs with less spoken content tend to achieve higher popularity.

4. **Speechiness Score:** Penalizes excessive spoken-word content.

$$\textbf{speechiness_score} = \begin{cases} 1, & \text{if } s \leq 0.10 \\ \max\left(0,\ 1 - \dfrac{s - 0.10}{0.40}\right), & \text{otherwise} \end{cases}$$

- Songs with less spoken content tend to achieve higher popularity.

5. **Instrumentalness Score:** Rewards tracks with vocals over purely instrumental compositions.

$$\textbf{instrumentalness_score} = \begin{cases} 1, & \text{if } i \leq 0.05 \\ \max\left(0,\ 1 - \dfrac{i - 0.05}{0.50}\right), & \text{otherwise} \end{cases}$$

- Mainstream songs are typically vocal-centric.

6. **Liveness Score:** Penalises live-recorded characteristics.

$$\textbf{liveness_score} = \begin{cases} 1, & \text{if } l \leq 0.20 \\ \max\left(0,\ 1 - \dfrac{l - 0.20}{0.60}\right), & \text{otherwise} \end{cases}$$

- Studio-recorded songs are generally more popular than live recordings

7. **Tempo Score:** Rewards songs within the optimal tempo range (80–130 BPM).

$$\textbf{tempo_score} = \begin{cases} 1, & 80 \leq t \leq 130 \\ \max\left(0,\ 1 - \dfrac{80 - t}{80}\right), & t < 80 \\ \max\left(0,\ 1 - \dfrac{t - 130}{130}\right), & t > 130 \end{cases}$$

- Songs within this tempo range align with listener preferences.

8. **Duration Score:** Measures closeness to ideal duration (~180 seconds).

$$\textbf{duration_score} = \max\left(0,\ 1 - \frac{|d_{ms} - 180000|}{180000}\right)$$

   - Songs of moderate length tend to retain listener attention.

9. **Loudness Alignment Score:** Evaluates alignment between loudness and energy.

$$\textbf{loudness_alignment_score} = \max\left(0,\ 1 - \left|e - \frac{\text{loudness} + 60}{55}\right|\right)$$

   - High-energy songs require corresponding loudness levels to maintain consistency.

10. **DNA Match Score:** All engineered feature scores are aggregated into a single metric:

$$\textbf{DNA Match Score} = \frac{1}{9}\sum_{k=1}^{9} \text{feature_score}_k$$

   - This score represents how closely a song follows the hypothesised "Hit Song DNA".
   - A higher DNA Match Score indicates stronger alignment with identified success patterns
   - A lower score suggests deviation from typical hit characteristics

- **Feature Consistency Ratio:** The Feature Consistency Ratio measures the proportion of features that strongly align with the "hit song DNA.

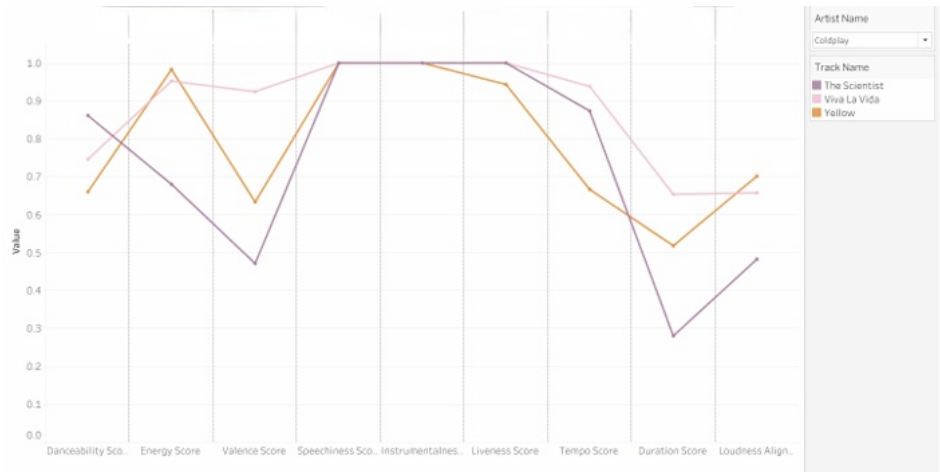
  Feature Variability = max(feature scores) − min(feature scores)

  - A high consistency ratio indicates that a song satisfies multiple optimal conditions Simultaneously, suggesting strong structural alignment with successful patterns.

- **Feature Variability:** Feature Variability measures the spread between the highest and lowest feature scores within a song.

  - Lower variability indicates balanced feature alignment, while higher variability suggests inconsistency across musical attributes. Songs with extreme variation may lack coherence in their overall structure.
  - This transformation enables a structured evaluation of songs beyond raw feature analysis and forms the foundation for the analysis in Hypothesis 4.

Q11: Feature alignment influences popularity with in an artist



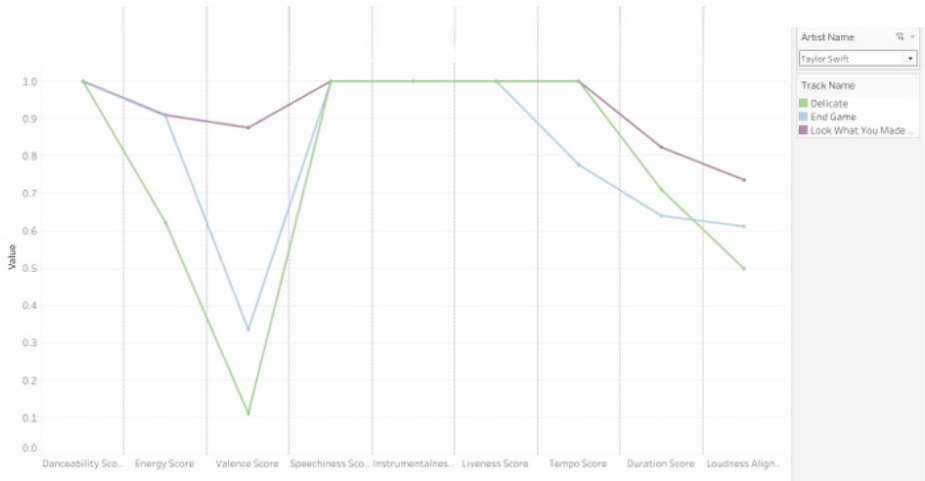


Figure 11. Artist-Level Feature Alignment Patterns

The line chart compares top songs from different artists based on engineered feature scores representing the "hit song DNA," where each feature is normalised to 0-1 to reflect alignment with optimal characteristics. While certain features, such as speechiness and instrumentalness, show consistently high alignment across songs, significant variation is observed in others, such as valence, duration, and loudness. This indicates that although successful songs tend to maintain a baseline structure for accessibility, they do not strictly adhere to a fixed feature profile. The variation across songs by the same artist suggests that popularity is not solely determined by perfect feature alignment but is also influenced by artistic style, creativity, and external factors. Overall, the analysis highlights that feature alignment is important but not sufficient, reinforcing the idea that success emerges from a balance between structured patterns and artist-driven uniqueness

Q12: Genre influences popularity distribution

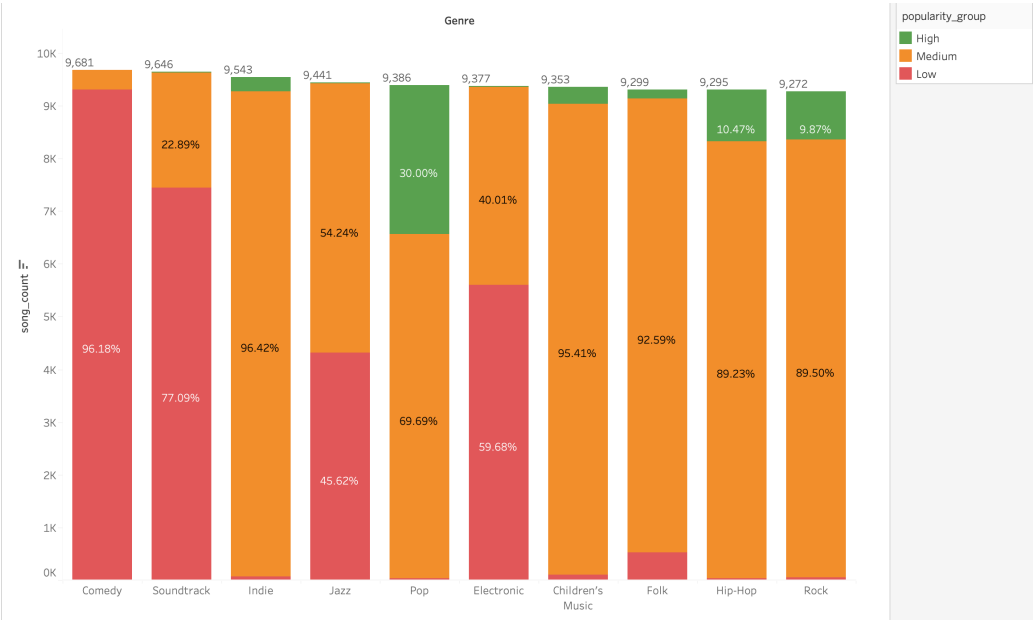


Figure 12. Popularity Distribution Across Genres

The stacked bar chart highlights clear differences in how popularity is distributed across genres. Most genres, including Indie, Children's Music, Folk, Hip-Hop, and Rock, are heavily concentrated in the medium popularity range, indicating stable but moderate audience reach. Comedy and Soundtrack stand out with a significantly higher share of low-popularity tracks, suggesting inconsistent performance and limited mainstream appeal. In contrast, Pop shows a more balanced distribution with a notable presence of high-popularity songs, reinforcing its strong connection with mass audiences.

Overall, the visual indicates that genre plays a meaningful role in shaping popularity patterns, particularly in determining consistency and likelihood of higher success. However, since no genre exclusively dominates the high-popularity segment, it also suggests that additional external factors beyond genre also influence a song's success.

Q13: Artist tier affects feature dependency

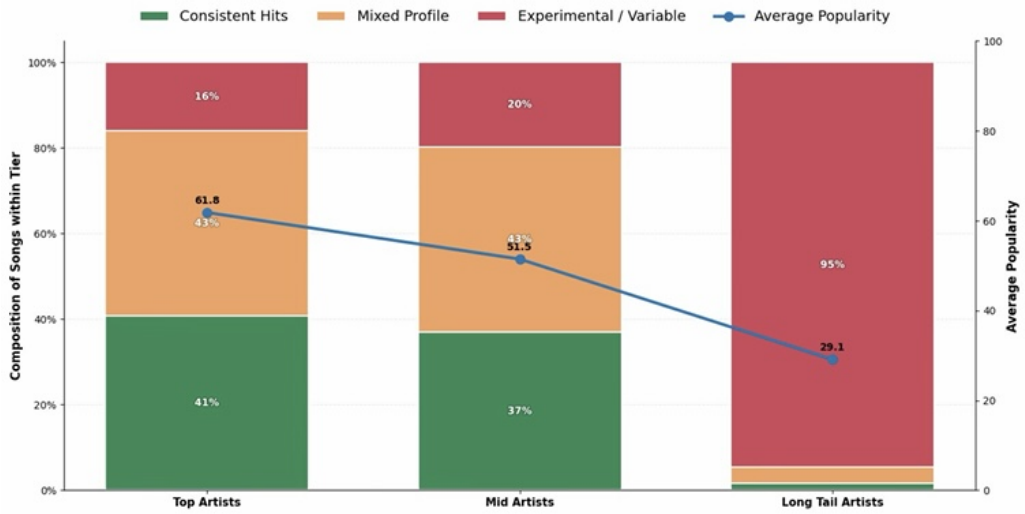


Figure 13. Artist Tier Composition vs Average Popularity

Top artists succeed with both consistency and flexibility, leveraging strong external influence to maintain high popularity even when deviating from optimal features. Mid-tier artists show moderate consistency but lack the same level of impact and audience reach. Long-tail artists, instead of following optimal feature patterns, lean heavily into experimentation in an effort to break through, but without external support, this strategy rarely translates into high popularity. This highlights a structural gap where experimentation alone cannot compensate for limited visibility and industry backing. Ultimately, success is driven by a combination of feature alignment and artist influence, not just one in isolation, and the ability to convert experimentation into reach.

Q14: External influence drives popularity beyond feature quality

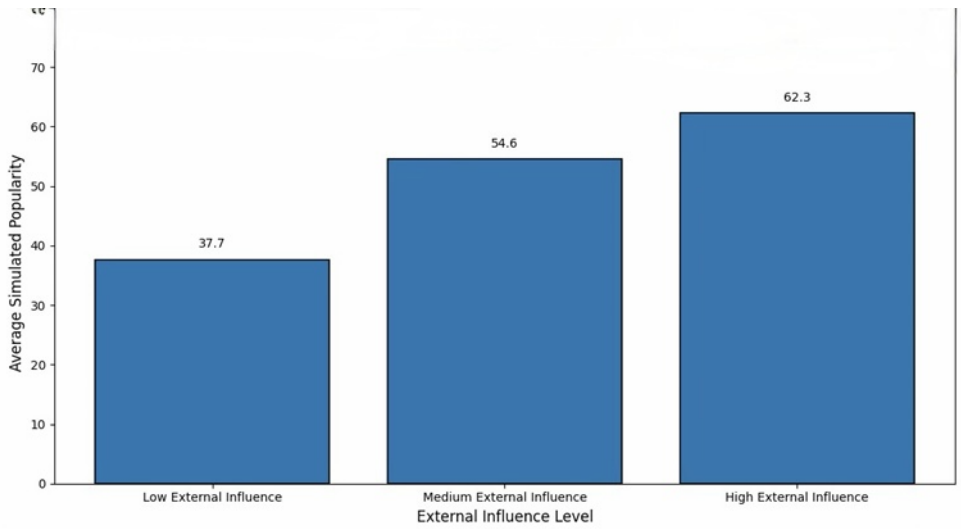


Figure 14. Impact of External Influence on Low-DNA Songs

To evaluate the role of external influence, a set of synthetic features was introduced to approximate real-world amplification factors that are not directly available in the dataset. These engineered variables simulate the effects of marketing exposure, social media virality, and platform-driven promotion, allowing us to quantify how non-audio factors contribute to a song's success. Collectively, these synthetic features constitute an aggregated External Influence Score that captures the overall impact of visibility, reach, and industry support on song popularity. The bar chart examines the impact of external influence on the popularity of songs with low DNA match scores, representing tracks that do not strongly align with the engineered "hit song" feature profile. Despite suboptimal feature alignment, a clear upward trend is observed as external influence increases, with average popularity rising significantly from low to high levels of influence. This demonstrates that external factors such as marketing, virality, and platform exposure can substantially enhance a song's performance, even when its intrinsic audio characteristics are weak. The findings highlight that popularity is not solely driven by feature quality, but is strongly amplified by external support mechanisms. In essence, external influence acts as a compensatory force, enabling songs with weaker structural attributes to achieve competitive levels of success

Narrative: This phase demonstrates that even well-aligned musical features are not sufficient to guarantee success. While the constructed "hit song DNA" captures important structural patterns, external influences such as artist popularity, genre trends, and platform-driven exposure play a crucial role in amplifying song reach. The results highlight that popularity emerges from a combination of intrinsic musical quality and external support mechanisms, completing the transition from feature-driven analysis to real-world dynamics.

Conclusion: Decoding the DNA of a Hit Song

This investigation shows that song popularity cannot be explained by a single factor. Instead, it is shaped by a combination of musical features and external influences.

The analysis highlights that individual audio features on their own do not strongly determine popularity. Songs that perform well usually have balanced characteristics and good alignment between multiple features, suggesting that overall musical harmony matters more than focusing on just one element.

However, even well-composed songs do not always guarantee success. External factors such as genre trends, artist popularity, platform promotion, and social media virality play a major role in increasing a song's reach and success. In many cases, these factors allow songs with poorer musical qualities to still become highly popular.

In conclusion, the DNA of a hit song is not only about how it is created, but also about how it is promoted, distributed, and discovered in today's music ecosystem.